5G RAN

Admission Control Feature Parameter Description

Issue Draft B

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised the definition of level-3 threshold UE in admission guarantee for voice service UEs. For details, see **Table 4-1**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-09-26).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added admission guarantee for voice service UEs. For details, see: • **4.1.1.1 Checking Whether UE Admission Is Limited** • **4.1.1.2.2 Checking Whether the Temporary RRC Connection Is Successfully Set Up for a UE** • **4.2.1 Benefits** • **4.2.2 Impacts** • **4.3.2.1 Prerequisite Functions** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added parameters: **NRCellPri**.***VoiceUeNumOfs*** **NRCellPri**.***TempAdmTimer*** **NRCellPri**.***TempAdmUeNumThld***	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the configuration of the admission threshold for emergency UEs performing an RRC initial access. For details, see **Table 4-1** in **4.1.1.1 Checking Whether UE Admission Is Limited**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the mutually exclusive relationship between service admission and user-rate-based dynamic network slice resource management. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the mutually exclusive relationship between service admission and user-rate-based dynamic QCI resource management. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
  - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-051102	Access Control	**4 Admission Control**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Admission control	Supported in both NSA and SA networking, with the following differences: • Network slice-based service admission and service preemption during bearer setup are not supported in NSA networking. • Service admission and service preemption for MCPTT and VoNR UEs are not supported in NSA networking.	**4 Admission Control**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

<table>
<tr><th>Function Name</th><th>Difference</th><th>Chapter/Section</th></tr>
<tr><td>Admission control</td><td>Supported in both high and low frequency bands, with the following differences:
UE admission: supported in both high and low frequency bands
UE preemption, service admission, and service preemption: supported only in low frequency bands
Service admission and service preemption for MCPTT and VoNR UEs: supported only in low frequency bands</td><td>4 Admission Control</td></tr>
</table>

3 Overview

When network congestion limits network capacity, the access success rate and handover success rate decrease. Admission control helps improve experience of high-priority UEs and services by increasing their access success rates and handover success rates. The admission control procedure mainly includes admission decision and preemption handling:

- Admission decision: The gNodeB determines whether to admit a UE based on the number of admitted UEs and licensed number of UEs. Additionally, the gNodeB determines whether to admit guaranteed bit rate (GBR) or non-GBR services based on the physical resource block (PRB) usage.
- Preemption handling: High-priority UEs or services preempt resources from low-priority UEs or services.

NOTE

Admission control consists of transmission-resource-based and radio-resource-based admission control. This document describes radio-resource-based admission control. For details about transmission-resource-based admission control, see *Transmission Resource Management*.

4 Admission Control

4.1 Principles

Admission control consists of UE admission control and service admission control.

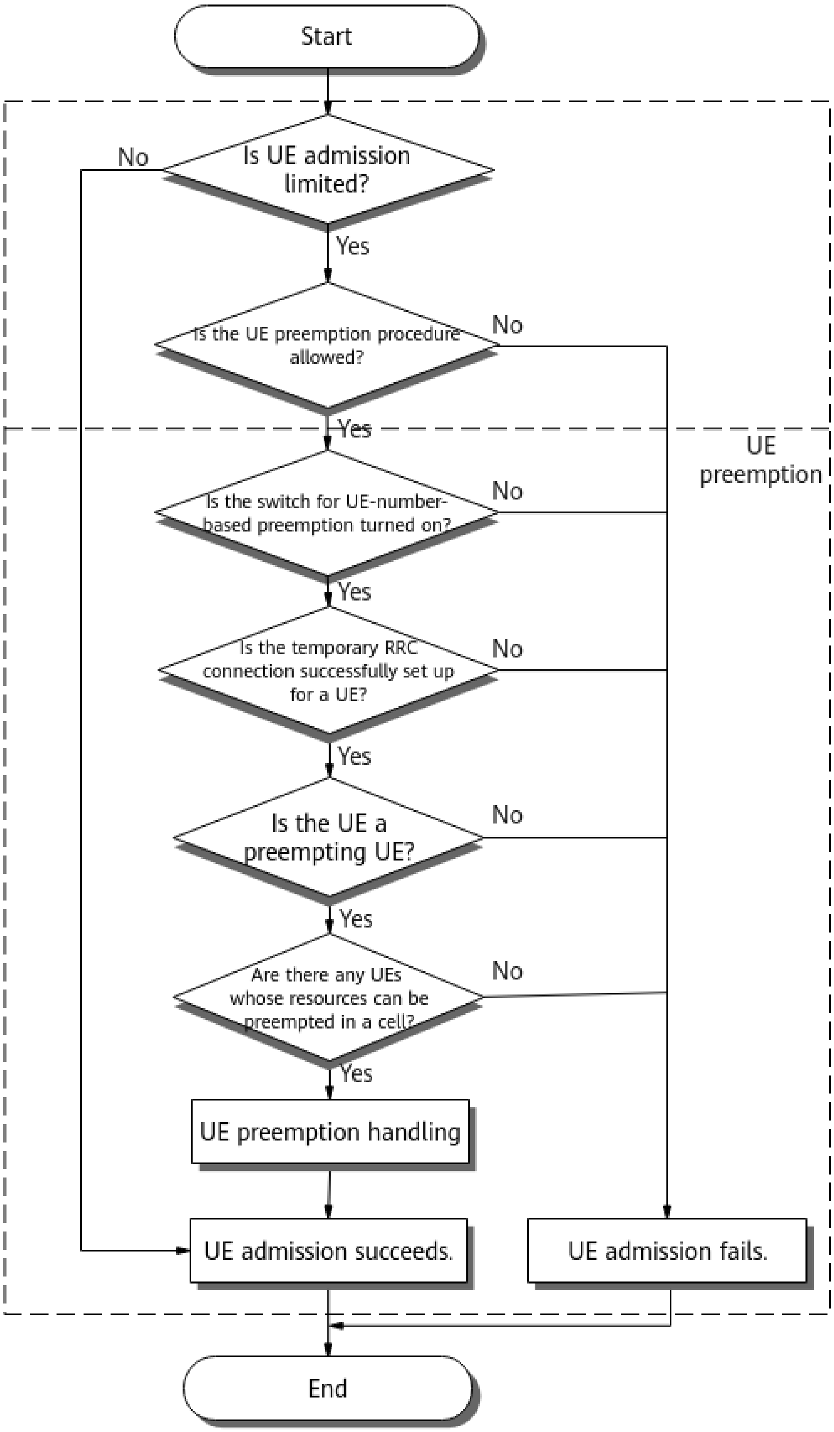
- When the licensed number of UEs is insufficient or the number of admitted UEs reaches the upper limit, UE admission control is performed on new UE access requests to ensure that the number of admitted UEs does not exceed the system specifications. UE admission control involves UE admission decision and UE preemption.
- When PRB resources or network slice resources in a cell are insufficient, service admission control is performed on new service access requests to ensure the QoS satisfaction rate of admitted UEs. Service admission control involves service admission and service preemption.

4.1.1 UE Admission Control

After a UE initiates a network access request, UE admission control is performed on the UE according to the procedure shown in **Figure 4-1**.

UE admission decision is supported in both high and low frequency bands. UE preemption is supported only in low frequency bands.

Figure 4-1 UE admission control procedure



Basic procedure of admission control:

1. **4.1.1.1 Checking Whether UE Admission Is Limited**:
   a. The gNodeB checks whether UE admission is limited. If not, the gNodeB determines that UE admission succeeds. If yes, the gNodeB proceeds to **b**.
   b. The gNodeB determines whether to enter the UE preemption procedure.

i. If UE admission is limited due to insufficient NR0S000SPE00 RE(Resource Element) License (NR) in the gNodeB, the gNodeB determines that UE admission fails.

ii. If UE admission is limited due to other causes, the gNodeB proceeds to **2**.

2. **4.1.1.2.1 Checking Whether the UE-Number-based Preemption Switch Is Turned On**

   If the UE-number-based preemption switch is turned off, the gNodeB determines that UE admission fails. If the switch is turned on, the gNodeB proceeds to **3**.

3. **4.1.1.2.2 Checking Whether the Temporary RRC Connection Is Successfully Set Up for a UE**:

   The gNodeB checks whether the temporary RRC connection is successfully set up. If not, the gNodeB determines that UE admission fails. If yes, the gNodeB proceeds to **4**.

4. **4.1.1.2.3 Checking Whether a UE Is a Preempting UE**

   The gNodeB checks whether a UE is a preempting UE. If not, the gNodeB determines that UE admission fails. If yes, the gNodeB proceeds to **5**.

5. **4.1.1.2.4 Checking Whether There Are UEs Whose Resources Can Be Preempted in a Cell**

   The gNodeB checks whether there are UEs whose resources can be preempted in the cell. If not, the gNodeB determines that UE admission fails. If yes, the gNodeB proceeds to **6**.

6. **4.1.1.2.5 Handling UE Preemption**

   Once preemption handling is complete, UE admission succeeds and the admission control procedure ends.

4.1.1.1 Checking Whether UE Admission Is Limited

When a new UE attempts to access the network, the gNodeB first checks whether UE admission is limited. The gNodeB determines that UE admission is limited if any of the following conditions is met:

- The NR0S0RRCUL00 RRC Connected User License (NR) in the gNodeB is insufficient.
- The NR0S000SPE00 RE(Resource Element) License (NR) in the gNodeB is insufficient.
- The number of UEs in a cell (including the admitted UEs of all types and UEs of the type that are accessing the network) reaches the upper limit. **Table 4-1** lists the admission thresholds for UEs of different types.

If UE admission is limited, the following rules apply:

- If UE admission is limited due to insufficient NR0S000SPE00 RE(Resource Element) License (NR) in the gNodeB, the gNodeB determines that UE admission fails.
- If UE admission is limited due to other causes, the gNodeB proceeds to **4.1.1.2.1 Checking Whether the UE-Number-based Preemption Switch Is Turned On**.

If UE admission is not limited, the gNodeB determines that UE admission succeeds.

Insufficient UE number resources due to license or base station configuration restrictions can degrade user experience for both calling and called parties in MCPTT and VoNR services. To address this issue, admission guarantee for voice services is introduced. It ensures that voice service UEs (level-3 threshold UEs in **Table 4-1**) are successfully admitted and can preempt resources from low-priority UEs. This function is enabled by setting the **NRCellPri**.***VoiceUeNumOfs*** parameter to a non-zero value. It applies only to low frequency bands.

Table 4-1 Four levels of UE-number-based admission thresholds

UE Type	UE Type Definition	Admission Threshold

Level-4 threshold UE	Any of the following conditions is met:
	• UE performing an intra-base-station incoming handover, with the cause value in the RRCSetupRequest message being **Emergency** or **highPriorityAccess**[a]
	• UE transiting from the inactive state to the connected state, with the cause value in the RRCResumeRequest1 message being **Emergency** or **highPriorityAccess**
	• UE performing an RRC initial access, with the cause value in the RRCSetupRequest message being **Emergency**[b] or **highPriorityAccess**
	• Emergency call UE[c] in the following scenarios:
	  – Inter- or intra-base-station incoming handover
	  – Transition from the inactive state to connected state
	  – RRC connection reestablishment
		The smaller of the following two values is used as the level-4 admission threshold:
		• **NRCellPri**.***UserNumAdmThld*** + **NRCellPri**.***HoReestUeNumOfs*** + **NRCellPri**.***VoiceUeNumOfs*** + **NRCellPri**.***HighPriUeNumOfs***
		• Maximum number of UEs that can access a cell under the limitation of hardware capabilities[d]

UE Type	UE Type Definition	Admission Threshold
Level-3 threshold UE	Any of the following conditions is met: ● UE involved in an inter-base-station incoming handover or intra-base-station incoming handover, with a 5QI 1/65/66 bearer established ● UE transiting from the inactive state to the connected state, with the cause value in the RRCResumeRequest1 message being **MO-VoiceCall** ● UE performing an RRC initial access, with the cause value in the RRCSetupRequest message being **MO-VoiceCall** ● UE performing an RRC connection reestablishment, with a 5QI 1/65/66 bearer established ● UE having a bearer with 5QI 1/65/66[e] set up before the timer specified by **NRCellPri**.***TempAdmTimer*** expires, with a cause value other than **Emergency**, **highPriorityAccess**, and **MO-VoiceCall**	The smaller of the following two values is used as the level-3 admission threshold: ● **NRCellPri**.***UserNumAdmThld*** + **NRCellPri**.***HoReestUeNumOfs*** + **NRCellPri**.***VoiceUeNumOfs*** ● Maximum number of UEs that can access a cell under the limitation of hardware capabilities[d]
Level-2 threshold UE	Any of the following conditions is met: ● Non-level-4 or non-level-3 threshold UE involved in an inter-base-station incoming handover ● Non-level-4 or non-level-3 threshold UE involved in an intra-base-station incoming handover ● Non-level-4 or non-level-3 threshold UE that transits from the inactive state to the connected state ● Non-level-4 or non-level-3 threshold UE involved in an RRC connection reestablishment	The smaller of the following two values is used as the level-2 admission threshold: ● **NRCellPri**.***UserNumAdmThld*** + **NRCellPri**.***HoReestUeNumOfs*** ● Maximum number of UEs that can access a cell under the limitation of hardware capabilities[d]

UE Type	UE Type Definition	Admission Threshold
Level-1 threshold UE	Common UE (not level-4, level-3, or level-2 threshold UE)	The smaller of the following two values is used as the level-1 admission threshold: ● **NRCellPri**.***UserNumAdmThld*** ● Maximum number of UEs that can access a cell under the limitation of hardware capabilities[d]

a: During an inter-base-station handover, the target cell cannot inherit the access cause value from the source cell, resulting in the failure to inherit the UE's threshold attributes. This may cause a level-4 threshold UE to become a level-2 threshold UE after the handover.

b: If a UE performs an RRC initial access with the cause value **Emergency** in the RRCSetupRequest message, its admission threshold is the smaller of the following two values: (1) the result of "**NRCellPri**.***TempAdmUeNumThld*** + **NRCellPri**.***UserNumAdmThld*** + **NRCellPri**.***HoReestUeNumOfs*** + **NRCellPri.*VoiceUeNumOfs*** + **NRCellPri**.***HighPriUeNumOfs***" and (2) the maximum number of UEs that can access a cell under the limitation of hardware capabilities[d].

c: Emergency call UEs are those for which the **gNBOperatorParam**.***EmcBearerArp*** parameter is set to the emergency call ARP configured on the core network when the **EMERGENCY_CALL_UE_ARP_IDENT_SW** option of the **gNB5qiConfig**.***Nr5qiAlgoSwitch*** parameter is selected.

d: For details about the maximum number of UEs that can access a cell under the limitation of hardware capabilities, see *BBU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

e: If admission guarantee for voice service UEs is enabled (by setting the **NRCellPri**.***VoiceUeNumOfs*** parameter to a non-zero value) for a cell, and the number of UEs in the cell is greater than or equal to the value of **NRCellPri**.***UserNumAdmThld***, then the UE with an access cause value other than **Emergency**, **highPriorityAccess**, and **MO-VoiceCall** is processed as follows:

- If the number of temporarily admitted UEs is less than the value of **NRCellPri**.***TempAdmUeNumThld***, the UE is temporarily admitted and a temporary RRC connection is set up. Before the timer specified by **NRCellPri**.***TempAdmTimer*** expires, the following logic applies:
  - If a bearer with 5QI 1/65/66 is set up for the UE, the UE is identified as a voice service UE and the level-3 admission threshold is applied.
  - If no such bearer is set up, the level-1 admission threshold is applied.
- If the number of temporarily admitted UEs is greater than or equal to the value of **NRCellPri.*TempAdmUeNumThld***, the temporary RRC connection fails to be set up and the UE admission fails.

NOTE

Both emergency call UEs and common UEs occupy the UE number resources of a baseband processing unit. To guarantee utilization of the UE number resources by common UEs when the UE number resources of the baseband processing unit are limited, "no resource reservation for emergency call UEs" is introduced. This function is controlled by the **RES_UNRESERVED_FOR_EMC_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter. If this option is selected, emergency call UEs and common UEs share the UE number resources that are not occupied on the baseband processing unit. If this option is deselected, common UEs cannot use the reserved UE number resources, reducing the UE number resources available to common UEs.

4.1.1.2 UE Preemption

4.1.1.2.1 Checking Whether the UE-Number-based Preemption Switch Is Turned On

UE-number-based preemption takes effect only in low frequency bands, and it is controlled by the **USER_NUM_PREEMPTION_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter. Before making UE admission decisions based on this option, the gNodeB determines whether a UE is an emergency call UE by checking the cause value in the RRCSetupRequest message, as follows:

- If the cause value is **Emergency**, the gNodeB proceeds to **4.1.1.2.2 Checking Whether the Temporary RRC Connection Is Successfully Set Up for a UE**.
- If the cause value is not **Emergency**, the gNodeB makes UE admission decisions based on the setting of the option that controls UE-number-based preemption, as follows:
  - If this option is deselected, the gNodeB determines that UE admission fails.
  - If this option is selected, the gNodeB proceeds to **4.1.1.2.2 Checking Whether the Temporary RRC Connection Is Successfully Set Up for a UE**.

In high frequency bands, the gNodeB determines that UE admission fails.

4.1.1.2.2 Checking Whether the Temporary RRC Connection Is Successfully Set Up for a UE

- When the **NRCellPri**.***VoiceUeNumOfs*** parameter is set to **0**, the following logic applies:

  In a UE preemption procedure, the gNodeB first attempts to set up a temporary RRC connection, through which the gNodeB expects to obtain a UE's allocation and retention priority (ARP) attributes (required in UE preemption decision-making).
  - If the temporary RRC connection is successfully set up, the gNodeB obtains the ARP attributes of the first bearer set up for the UE within 5s.
    - If the UE's ARP attributes are successfully obtained within 5s, the gNodeB promptly proceeds to **4.1.1.2.3 Checking Whether a UE Is a Preempting UE**.

        - The gNodeB determines that UE admission fails if the UE's ARP attributes fail to be obtained within 5s.
    - If the temporary RRC connection fails to be set up, UE admission fails.
- When the **NRCellPri**.***VoiceUeNumOfs*** parameter is set to a non-zero value, the following logic applies:
    - If the access cause value is **Emergency**, **highPriorityAccess**, or **MO-VoiceCall** for a UE, the gNodeB attempts to set up a temporary RRC connection with the UE to obtain the UE's ARP attributes for user preemption decision.
        - If the temporary RRC connection is successfully set up, the gNodeB obtains the ARP attributes of the first bearer set up for the UE within 5s.
            - If the UE's ARP attributes are successfully obtained within 5s, the gNodeB promptly proceeds to **4.1.1.2.3 Checking Whether a UE Is a Preempting UE**.
            - The gNodeB determines that UE admission fails if the UE's ARP attributes fail to be obtained within 5s.
        - If the temporary RRC connection fails to be set up, UE admission fails.
    - If the access cause value is not **Emergency**, **highPriorityAccess**, or **MO-VoiceCall** and a temporary RRC connection is set up for a UE, the gNodeB obtains the ARP attributes of the highest-priority bearer of the UE after the timer specified by **NRCellPri**.***TempAdmTimer*** expires.
        - If the ARP attributes are successfully obtained, the gNodeB promptly proceeds to **4.1.1.2.3 Checking Whether a UE Is a Preempting UE**.
        - If no bearer is set up for the UE after the timer specified by **NRCellPri**.***TempAdmTimer*** expires, the gNodeB determines that the UE admission fails.

If UE admission fails, the gNodeB immediately releases the temporary RRC connection. If UE admission succeeds, the temporary RRC connection is changed to a formal one.

NOTE

ARP attributes are defined in section 9.3.1.19 "QoS Flow Level QoS Parameters" in 3GPP TS 38.413 V15.4.0 and are separately represented by the following IEs:

- *Priority Level*: The IE value ranges from 1 to 15 with **1** as the highest level of priority. A high-ARP-priority UE can preempt the resources of a low-ARP-priority UE.
- *Pre-emption Capability*
    - If the IE value is **may trigger pre-emption**, the UE can preempt resources of low-priority UEs.
    - If the IE value is **shall not trigger pre-emption**, the UE cannot preempt resources of low-priority UEs.
- *Pre-emption Vulnerability*
    - If the IE value is **pre-emptable**, the resources allocated to the UE can be preempted by other UEs.
    - If the IE value is **not pre-emptable**, the resources allocated to the UE cannot be preempted by other UEs.

4.1.1.2.3 Checking Whether a UE Is a Preempting UE

The gNodeB considers a UE under limited admission as a preempting UE when all of the following conditions are met:

- The number of admitted UEs reaches the upper limit, and the UE is not an SCC CA UE and is not enabled with supplementary uplink (SUL).
- SA UE: The cause value in the RRCSetupRequest message is **Emergency**, **highPriorityAccess**, or **mo-VoiceCall**. Alternatively, the ARP priority (indicated by *Priority Level*) is not 15, and the value of *Pre-emption Capability* is **may trigger pre-emption**.

  NSA UE: The ARP priority (indicated by *Priority Level*) is not 15 and the value of *Pre-emption Capability* is **may trigger pre-emption**.

If a UE meets the preempting conditions, the gNodeB proceeds to **4.1.1.2.4 Checking Whether There Are UEs Whose Resources Can Be Preempted in a Cell**. If the UE does not meet preempting requirements, the gNodeB determines that UE admission fails.

4.1.1.2.4 Checking Whether There Are UEs Whose Resources Can Be Preempted in a Cell

Resources of a UE cannot be preempted if any of the following conditions is met:

- The cause value in the RRCSetupRequest message is **Emergency** or **highPriorityAccess**.
- The value of the *Pre-emption Vulnerability* IE corresponding to any bearer of a UE is **not pre-emptable**. When **NRDUCellRac**.***SpecServPviMode*** is set to **PVI_FOLLOW_ARP**, the preemption vulnerability attribute of a UE of any of the special UE types takes the *Pre-emption Vulnerability* value delivered by the core network.
- The **HIGH_ACCESS_PRIORITY** option of the **gNBOperatorQciParam**.***RacSwitch*** parameter corresponding to the QoS class identifier (QCI) of any bearer of the UE is selected.
- The **NRDUCellRac**.***SpecServPviMode*** parameter is set to **OFF** for a UE of any of the special UE types.
- The UE's PLMN is out of the PLMN range for preemption.

  The **NRDUCellRac**.***UeNumPreemptionRange*** parameter specifies the PLMN range for UE-number-based preemption.

  - If the **NRDUCellRac**.***UeNumPreemptionRange*** parameter is set to **INTRA_OPERATOR**, a preempted UE must have the same PLMN as the preempting UE.
  - If the **NRDUCellRac**.***UeNumPreemptionRange*** parameter is set to **INTER_OPERATOR**, a preempted UE must have a different PLMN from the preempting UE.
  - If the **NRDUCellRac**.***UeNumPreemptionRange*** parameter is set to **ALL_OPERATOR**, a preempted UE can have the same PLMN as or a different PLMN from the preempting UE.

  The **NRDUCellRac**.***UeNumPreemptionRange*** parameter can be set to **INTER_OPERATOR** or **ALL_OPERATOR** only when the **gNBSharingMode**.***gNBMultiOpSharingMode*** parameter is set to **SHARED_FREQ** (indicating RAN sharing with common carrier).

Other UEs are sorted in ascending order of ARP priority, and their resources can be preempted.

- If there are UEs whose resources can be preempted, the gNodeB proceeds to **4.1.1.2.5 Handling UE Preemption**.
- If there are no UEs whose resources can be preempted, the gNodeB determines that UE admission fails.

NOTE

The special UE types include:

- Voice service UE (with the cause value in the RRCSetupRequest message being **mo-VoiceCall**)
- UE performing services with low-latency requirements
- SUL UE
- Super uplink UE
- FWA private line UE

4.1.1.2.5 Handling UE Preemption

The preempting UE proceeds to the subsequent random access, incoming handover, or RRC connection reestablishment procedure.

The processing for UEs whose resources are pre-emptable differs between SA and NSA as follows:

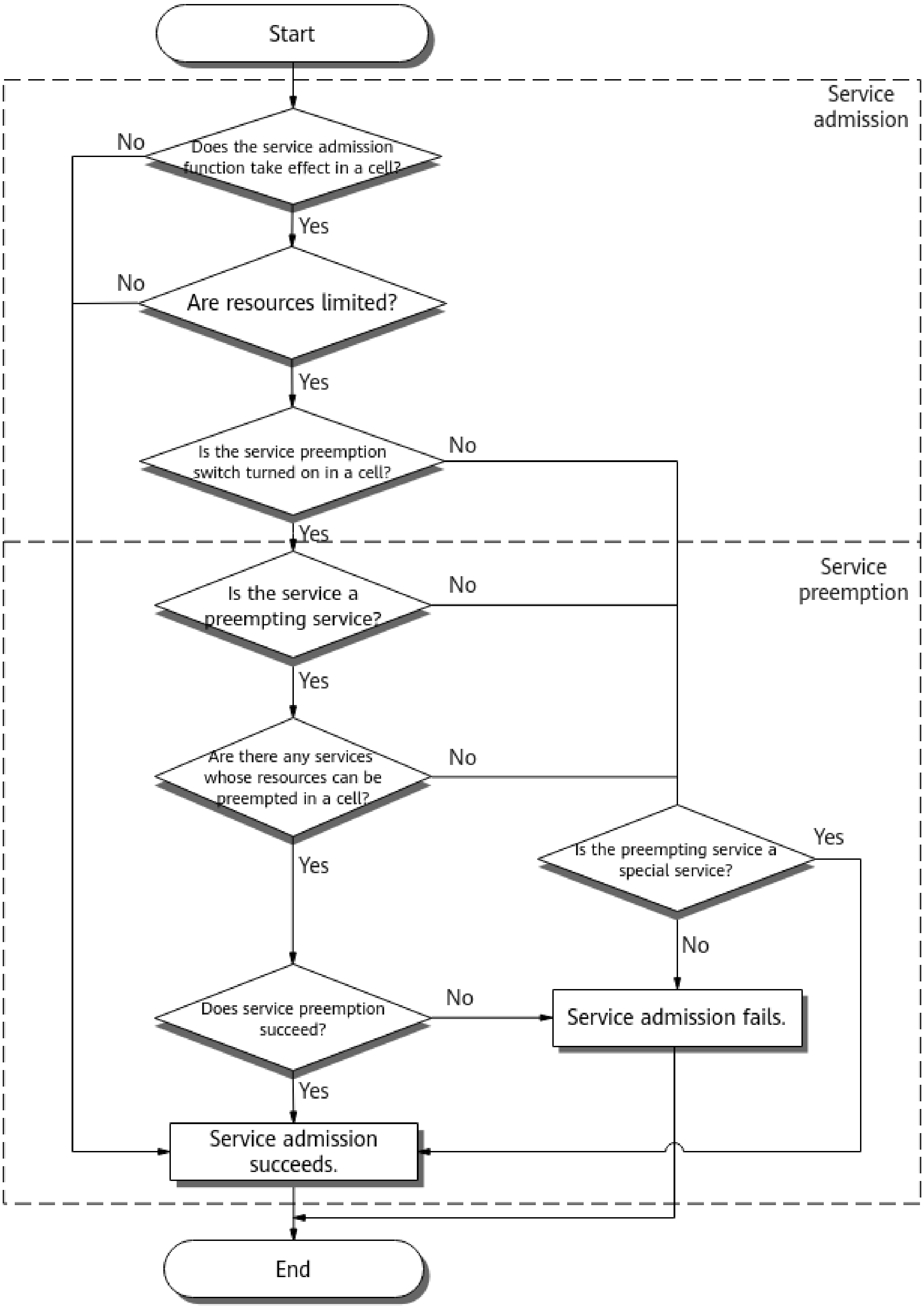
- For an SA UE, the gNodeB delivers an RRC Release message to release the UE and sends a UE CONTEXT RELEASE REQUEST message with the cause value of **Release due to pre-emption** over the NG interface.
  - If the **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected, the gNodeB includes the *RejectWaitTime* IE in the RRC Release message, instructing the UE to initiate the random access, incoming handover, or RRC connection reestablishment procedure 16s later.
  - If the **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter is deselected, the gNodeB does not include the *RejectWaitTime* IE in the RRC Release message, and the UE can initiate the random access, incoming handover, or RRC connection reestablishment procedure at any time.
- For an NSA UE, the gNodeB releases the secondary gNodeB (SgNB) connection with the cause value of **Reduce Load**.

4.1.2 Service Admission Control

After a service access request is initiated, service admission control is performed according to the procedure shown in **Figure 4-2**.

Service admission control involves service admission and service preemption, which take effect only in low-frequency scenarios.

Figure 4-2 Service admission control procedure



Basic procedure of service admission:

1. **4.1.2.1.1 Checking Whether the Service Admission Function Takes Effect in a Cell**

   The gNodeB checks whether service admission takes effect in a cell. If yes, the gNodeB proceeds to 2. If not, the gNodeB determines that service admission succeeds.

2. **4.1.2.1.2 Checking Whether Resources Are Limited**

   If cell resources are not limited, the gNodeB determines that service admission succeeds. If cell resources are limited, the gNodeB proceeds to **3**.

3. **4.1.2.2.1 Checking Whether the Service Preemption Switch Is Turned On in a Cell**

   If the service preemption switch is turned off, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**.

   – If the preempting service is a special service, the gNodeB determines that service admission succeeds.

   – If the preempting service is not a special service, the gNodeB determines that service admission fails.

   If the service preemption switch is turned on, the gNodeB proceeds to **4**.

4. **4.1.2.2.2 Checking Whether a Service Is a Preempting Service**

   For a service that does not meet the preempting conditions, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**. For a service that meets the preempting conditions, the gNodeB proceeds to **5**.

5. **4.1.2.2.3 Checking Whether There Are Services Whose Resources Can Be Preempted in a Cell**

   If there are no services whose resources can be preempted, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**. If there are services whose resources can be preempted, the gNodeB proceeds to **6**.

6. **4.1.2.2.5 Handling Service Preemption**

   If service preemption handling succeeds, service admission succeeds and the service admission procedure ends. If service preemption handling fails, service admission fails.

4.1.2.1 Service Admission

4.1.2.1.1 Checking Whether the Service Admission Function Takes Effect in a Cell

Service admission is controlled by the **QOS_FLOW_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter. The gNodeB makes service admission decisions based on the setting of this option and other conditions required for the function to take effect. Further details are as follows:

- If this option is selected and all other required conditions are met, the gNodeB determines that service admission takes effect in the cell and then the gNodeB proceeds to **4.1.2.1.2 Checking Whether Resources Are Limited**. The aforementioned conditions are as follows:
  - The average number of RRC_CONNECTED UEs in the cell (indicated by **N.User.RRCConn.Avg**) exceeds 20.
  - The uplink PRB usage is greater than 20% or the downlink PRB usage is greater than 10%. The PRB usage is calculated as follows:
    - Uplink PRB usage = **N.PRB.UL.Used.Avg**/Maximum number of PRBs in a cell
    - Downlink PRB usage = **N.PRB.DL.Used.Avg**/Maximum number of PRBs in a cell

- In other cases, service admission does not take effect in the cell, and services can be directly admitted.

4.1.2.1.2 Checking Whether Resources Are Limited

Admission of services carried on the bearers of SCC CA UEs or SUL UEs can be directly admitted without resource limitation check because these services are not counted into the load of the cell.

For other services, the gNodeB checks whether cell resources are limited. If a UE is an SA UE and the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter is selected, the gNodeB also checks whether slice resources are limited.

- For details about how to check whether cell resources are limited, see **Checking Whether Cell Resources Are Limited**.
- For details about how to check whether slice resources are limited, see **Checking Whether Slice Resources Are Limited**.

If resources are not limited, the gNodeB determines that service admission succeeds. If resources are limited, the gNodeB proceeds to **4.1.2.2.1 Checking Whether the Service Preemption Switch Is Turned On in a Cell**.

Checking Whether Cell Resources Are Limited

For all SA and NSA UEs, cell resources are not limited when both of the following conditions are met. Otherwise, cell resources are limited.

- (Sum of downlink PRB usages of online UEs in a cell + Downlink PRB usage of the bearer to be admitted) ≤ Downlink PRB usage threshold of a cell
- (Sum of uplink PRB usages of online UEs in a cell + Uplink PRB usage of the bearer to be admitted) ≤ Uplink PRB usage threshold of a cell

The sum of uplink/downlink PRB usages of online UEs in a cell is estimated based on the uplink/downlink guaranteed rates of admitted services in the cell and the spectral efficiency of the cell under ideal conditions. The uplink and downlink guaranteed rates are indicated or configured as follows:

- For GBR services, the uplink and downlink guaranteed rates are indicated to the gNodeB by the core network through the *Guaranteed Flow Bit Rate Uplink* and *Guaranteed Flow Bit Rate Downlink* IEs, respectively. When the uplink/downlink guaranteed rate is increased on the core network, service admission may be triggered again. However, if the cell load is high, the admission will fail, leading to a service release.
- For non-GBR services, the uplink and downlink guaranteed rates are configured on the gNodeB by using **gNBDUMacParamGroup**.***UlGuaranteedRate*** and **gNBDUMacParamGroup**.***DlGuaranteedRate***, respectively. When the parameter values are changed to larger ones, service admission may be triggered again. However, if the cell load is high, the admission will fail, leading to a service release.

Checking Whether Slice Resources Are Limited

- When the bearer carrying the service to be admitted belongs to a network slice group *N*, slice resources will be limited if either of the following conditions is not met:
  - (Downlink PRB usage of the service to be admitted + Sum of downlink PRB usages of online UEs in network slice group *N*) ≤ Downlink bearer admission threshold of network slice group *N*
  - (Uplink PRB usage of the service to be admitted + Sum of uplink PRB usages of online UEs in network slice group *N*) ≤ Uplink bearer admission threshold of network slice group *N*
- When the bearer carrying the service to be admitted does not belong to any network slice group, slice resources will be limited if either of the following conditions is not met:
  - (Sum of downlink PRB usages of online UEs outside network slice groups + Downlink service PRB usage) ≤ Non-network-slice-group downlink bearer admission threshold
  - (Sum of uplink PRB usages of online UEs outside network slice groups + Uplink service PRB usage) ≤ Non-network-slice-group uplink bearer admission threshold

Table 4-2 lists the parameters for checking whether slice resources are limited.

Table 4-2 Parameters for checking whether slice resources are limited

Parameter	Definition
Uplink/Downlink PRB usage of a to-be-admitted service	Estimated based on the guaranteed rate of the service and the spectral efficiency under ideal cell conditions.
Sum of uplink/downlink PRB usages of online UEs in network slice group *N*	Estimated based on the sum of admitted services' guaranteed rates and the spectral efficiency under ideal cell conditions.

Parameter	Definition
Downlink bearer admission threshold of network slice group *N*	Downlink bearer admission threshold of network slice group *N* = Downlink PRB usage threshold of the cell x Min{**NRDUCellRes**.***DlRbMaxRatio*** parameter value (in the unit of %) of network slice group *N*, (1 – Downlink accumulated sum)} Note: 1. Downlink PRB usage threshold of the cell = **NRDUCellRac**.***DlPrbUsageThld*** x (1 + **NRDUCellRac**.***DlAdmissionThldFactor***/10) 2. If **NRDUCellRes**.***DlRbMaxRatio*** is not configured, then the value **100** is used as the **NRDUCellRes**.***DlRbMaxRatio*** parameter value in the preceding formula. 3. The downlink accumulated sum can be 0% or the sum of the **NRDUCellRes**.***DlRbAverageRatio*** or **NRDUCellRes**.***DlRbMinRatio*** parameter values of other network slice groups, depending on the following rules: a. For any of other network slice groups, if both **NRDUCellRes**.***DlRbAverageRatio*** and **NRDUCellRes**.***DlRbMinRatio*** have valid configurations (set to a value ranging from **1** to **100**), the **NRDUCellRes**.***DlRbAverageRatio*** parameter value is used to calculate the downlink accumulated sum. b. For any of other network slice groups, if either **NRDUCellRes**.***DlRbAverageRatio*** or **NRDUCellRes**.***DlRbMinRatio*** has the valid configuration, the valid parameter configuration is used to calculate the downlink accumulated sum. c. For any of other network slice groups, if neither **NRDUCellRes**.***DlRbAverageRatio*** nor **NRDUCellRes**.***DlRbMinRatio*** has valid configurations, the downlink accumulated sum is 0%.

Parameter	Definition
Uplink bearer admission threshold of network slice group *N*	Uplink bearer admission threshold of network slice group *N* = Uplink PRB usage threshold of the cell x Min{**NRDUCellRes**.***UlRbMaxRatio*** parameter value (in the unit of %) of network slice group *N*, (1 – Uplink accumulated sum)} Note: 1. Uplink PRB usage threshold of the cell = **NRDUCellRac**.***UlPrbUsageThld*** x (1 + **NRDUCellRac**.***UlAdmissionThldFactor***/10) 2. If **NRDUCellRes**.***UlRbMaxRatio*** is not configured, then the value **100** is used as the **NRDUCellRes**.***UlRbMaxRatio*** parameter value in the preceding formula. 3. The uplink accumulated sum can be 0% or the sum of the **NRDUCellRes**.***UlRbAverageRatio*** or **NRDUCellRes**.***UlRbMinRatio*** parameter values of other network slice groups, depending on the following rules: a. For any of other network slice groups, if both **NRDUCellRes**.***UlRbAverageRatio*** and **NRDUCellRes**.***UlRbMinRatio*** have valid configurations (set to a value ranging from **1** to **100**), the **NRDUCellRes**.***UlRbAverageRatio*** parameter value is used to calculate the uplink accumulated sum. b. For any of other network slice groups, if either **NRDUCellRes**.***UlRbAverageRatio*** or **NRDUCellRes**.***UlRbMinRatio*** has the valid configuration, the valid parameter configuration is used to calculate the uplink accumulated sum. c. For any of other network slice groups, if neither **NRDUCellRes**.***UlRbAverageRatio*** nor **NRDUCellRes**.***UlRbMinRatio*** has valid configurations, the uplink accumulated sum is 0%.
Non-network-slice-group downlink bearer admission threshold	Non-network-slice-group downlink bearer admission threshold = Downlink PRB usage threshold of the cell x (1 – Downlink accumulated sum) The downlink accumulated sum can be the sum of the **NRDUCellRes**.***DlRbAverageRatio*** or **NRDUCellRes**.***DlRbMinRatio*** parameter values of network slice groups.
Non-network-slice-group uplink bearer admission threshold	Non-network-slice-group uplink bearer admission threshold = Uplink PRB usage threshold of the cell x (1 – Uplink accumulated sum) The uplink accumulated sum can be the sum of the **NRDUCellRes**.***UlRbAverageRatio*** or **NRDUCellRes**.***UlRbMinRatio*** parameter values of network slice groups.

NOTE

- If a UE is engaged with multiple bearers, some of which belong to network slice groups, then admission and preemption are performed for these bearers separately in and outside network slice groups. If a UE is configured with multiple network slice groups, then admission and preemption are performed on a per network slice group basis. For details about network slicing, see *Network Slicing*.
- Both non-network-slice-group admission failures and network-slice-group admission failures are caused with the same cause value **Resources not available for the slice(s)**, and are processed in the same way.

4.1.2.2 Service Preemption

4.1.2.2.1 Checking Whether the Service Preemption Switch Is Turned On in a Cell

Service preemption is controlled by the **QOS_FLOW_PREEMPTION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter. The gNodeB makes service preemption decisions based on the setting of this option.

- If this option is deselected, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**.
- If this option is selected, the gNodeB proceeds to **4.1.2.2.2 Checking Whether a Service Is a Preempting Service**.

If cell resources are insufficient, UE admission may fail, or the RRC connection of a UE may be released due to resource preemption. When this happens, the gNodeB instructs such UEs to delay access by delivering a message indicating waiting time of 16s if the **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected.

4.1.2.2.2 Checking Whether a Service Is a Preempting Service

The gNodeB considers a service under limited admission as a preempting service when the bearer carrying the service meets all of the following conditions:

- The bearer carries services of non-SCC CA UEs.
- The bearer carries services of non-SUL UEs.
- For the bearer, *Priority Level* is not **15** and *Pre-emption Capability* is set to **may trigger pre-emption**.

For a service that meets the preempting conditions, the gNodeB proceeds to **4.1.2.2.3 Checking Whether There Are Services Whose Resources Can Be Preempted in a Cell**. For a service that does not meet the preempting conditions, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**.

4.1.2.2.3 Checking Whether There Are Services Whose Resources Can Be Preempted in a Cell

In service preemption, all PRB resources and network slice resources of a UE, rather than those of a single bearer of the UE, are preempted. A UE's resources cannot be preempted when the bearer carrying any of its services meets any of the following conditions:

- The value of *Pre-emption Vulnerability* is **not pre-emptable** for the bearer, indicating that the bearer cannot be preempted.
- The **HIGH_ACCESS_PRIORITY** option of the **gNBOperatorQciParam**.***RacSwitch*** parameter corresponding to the QCI of the service bearer is selected.
- The bearer carries services that do not belong to the same PLMN as the preempting service.
- For GBR bearers, **Guaranteed Flow Bit Rate Uplink** and **Guaranteed Flow Bit Rate Downlink** are set to **0**.
- For non-GBR bearers, neither **gNBDUMacParamGroup**.***DlGuaranteedRate*** nor **gNBDUMacParamGroup**.***UlGuaranteedRate*** is configured, either of them is set to **0**, or both are set to **0**. Value **0** indicates that the admission rate is 0 kbit/s.
- When **NRDUCellRac**.***SpecServPviMode*** is set to **OFF** and any service of a UE is carried on a special bearer, the UE's resources cannot be preempted.
- When **NRDUCellRac**.***SpecServPviMode*** is set to **PVI_FOLLOW_ARP** and any service of a UE is carried on a special bearer, the UE's preemption vulnerability attribute takes the ARP attribute delivered by the core network.

Other UEs are sorted in ascending order of ARP priority, and their resources can be preempted. Among UEs of the same ARP priority, the resources of the UE whose total rate of all bearers is the highest are preferentially preempted.

- If there are UEs whose resources can be preempted, the gNodeB proceeds to **4.1.2.2.5 Handling Service Preemption**.
- If there are no UEs whose resources can be preempted, the gNodeB proceeds to **4.1.2.2.4 Checking Whether a Service Is a Special Service**.

NOTE

The special bearer types include:

- Bearer carrying services of UEs performing voice services
- Bearer carrying services of UEs performing services with low-latency requirements
- Bearer carrying services of SUL UEs
- Bearer carrying services of super uplink UEs
- Bearer carrying services of FWA private line UEs

4.1.2.2.4 Checking Whether a Service Is a Special Service

A special service is the service carried on a bearer for which the **HIGH_ACCESS_PRIORITY** option of the **gNBOperatorQciParam**.***RacSwitch*** parameter is selected.

- If the service to be admitted is a special service, the gNodeB determines that service admission succeeds.
- If the service to be admitted is not a special service, the gNodeB determines that service admission fails.

4.1.2.2.5 Handling Service Preemption

Service preemption is a process in which the preempting service preempts PRB resources of the service to be preempted.

- If the preemption is caused by insufficient network slice resources, the available PRB resources to be preempted are calculated based on the set of services whose resources can be preempted in network slice groups and those outside network slice groups.
- If the preemption is caused by insufficient PRB resources in a cell, the available PRB resources to be preempted are calculated based on the set of services that can be preempted in the cell.

If the number of PRBs released from preempted services meets the requirements of the preempting service, the preemption succeeds. Otherwise, the preemption fails. A preempting service can preempt the PRB resources of up to 10 preempted UEs. Details are as follows:

- If the UEs to be preempted can be found and the number of PRBs released from the found preempted UEs (fewer than 10 UEs) suffice for the preempting service, the gNodeB determines that service preemption succeeds.
  – If a UE to be preempted is an SA UE, the gNodeB delivers an RRC Release message that contains the *RejectWaitTime* IE and the cause value of **Release due to pre-emption**, instructing the UE to initiate network access 16s later. The **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter specifies whether the *RejectWaitTime* IE is contained in the message.
  – If a UE to be preempted is an NSA UE, the gNodeB releases the SgNB connection with the cause value of **Reduce Load**.
- If no UEs to be preempted are found or if 10 UEs to be preempted are found but the total number of PRBs released is insufficient for the preempting service, the gNodeB determines that service preemption fails.

NOTE

Making the following configuration changes during busy hours will lead to changes in service admission statistics:

- Turning on the service admission switch. When the service admission switch is turned on in a cell, the statistics of only newly admitted services, rather than the previously admitted services, are collected in the cell and network slice groups for the cell.
- Running the **ADD NRDUCELLRES** command. When this command is executed, the statistics of the previously admitted services are still collected in the original network slice group, but not in the new network slice group.
- Running the **RMV NRDUCELLRES** command. When this command is executed, the statistics of the services that have been admitted to the removed network slice group are not collected in other network slice groups.

The number of PRBs required by the preempting service can be estimated based on the spectral efficiency under ideal cell conditions and QoS parameters of the preempting service.

4.2 Network Analysis

4.2.1 Benefits

Admission control guarantees the QoS of admitted UEs by controlling admission requests of new UEs and services to maximize resource utilization. With this function, high-priority UEs are allowed to preempt resources of low-priority UEs, and likewise, high-priority services are allowed to preempt resources of low-priority services, improving experience of both high-priority UEs and services.

- The UE admission function is enabled by default.
- It is recommended that the UE preemption function be enabled when all of the following conditions are met:
  - The value of the **N.RRCConn.UserLic.Limit.Num.PLMN** counter (used to measure the number of times the licensed number of RRC_CONNECTED UEs is limited for a specific operator) is constantly greater than 0, or the value of the **N.User.RRCConn.Max** counter (used to measure the maximum number of UEs in RRC_CONNECTED mode in a cell) frequently reaches the upper limit of the hardware capability.
  - The value of the **N.UECntx.FailEst.NoRadioRes** counter (used to measure the number of UE context setup failures because of insufficient radio resources in a cell) or the **N.NsaDc.SgNB.Add.Fail.Radio.License** counter (used to measure the number of SgNB addition failures caused by user number or RE license insufficiency in LTE-NR NSA DC scenarios) is constantly greater than 0.
- It is recommended that the service preemption function be enabled when all of the following conditions are met:
  - Service admission has taken effect.
  - The number of QoS flow setup failures for a specific 5QI (**N.QosFlow.Est.Att.Cell5QI – N.QosFlow.Est.Succ.Cell5QI**) is constantly greater than 0 in a cell.

When none of the preceding conditions is met, enabling the admission control function offers fewer benefits than expected but does not have negative impacts.

When the **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected, the bearer setup success rate (**QoS Flow Setup Success Rate (CU)**) increases in scenarios where the cell/network slice load is relatively high.

When the **NRCellPri**.***VoiceUeNumOfs*** parameter is set to a non-zero value, the access success rate of voice service UEs increases.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- After four levels of UE-number-based admission thresholds are set, the number of UEs that fail to be admitted may increase when the admission threshold of any level is low.
- For SA UEs under heavy load:
  - After UE preemption takes effect, the values of **RRC Setup Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** increase. In addition, the number of abnormal RRC connection releases caused when the resources of UEs with low ARP priorities are preempted increases. As a result, the value of **Service Call Drop Rate (CU)** increases for QoS flows.

  - After service admission takes effect, the average rate of services with high ARP priorities increases but the value of **QoS Flow Setup Success Rate (CU)** decreases.
  - After service preemption takes effect, the setup success rate and average rate of services with high ARP priorities increase. However, the services of UEs with low ARP priorities are preempted, which increases the value of **Service Call Drop Rate (CU)** for QoS flows.
- For NSA UEs under heavy load:
  - After UE preemption takes effect, the following KPIs improve:
    - **Intra-SgNB PSCell Change Success Rate (CU)**, which indicates the intra-SgNB cell change success rate in NSA DC scenarios
    - **Intra-SgNB IntraFreq PSCell Change Success Rate (CU)**, which indicates the intra-SgNB intra-frequency cell change success rate in NSA DC scenarios
    - **Inter-SgNB PSCell Change Success Rate (CU)**, which indicates the inter-SgNB cell change success rate in NSA DC scenarios
    - **SgNB Addition Success Rate (CU)**, which indicates the SgNB addition success rate
    - **Inter-SgNB IntraFreq PSCell Change Success Rate (CU)**, which indicates the inter-SgNB intra-frequency cell change success rate in NSA DC scenarios
  - After UE preemption takes effect, the number of abnormal RRC connection releases caused when the resources of UEs with low ARP priorities are preempted increases, causing the following KPIs to increase:
    - **SgNB-Triggered SgNB Abnormal Release Rate (CU)**, which indicates the abnormal SgNB releases triggered by the SgNB in NSA DC scenarios
    - **SgNB Abnormal Release Rate (CU)**, which indicates the abnormal SgNB release rate in NSA DC scenarios
  - After service admission takes effect, the average rate of services with high ARP priorities increases. As a result, the number of successful data radio bearer (DRB) additions for LTE-NR NSA DC UEs on the SgNB (**N.NsaDc.DRB.Add.Succ**) decreases.
  - After service preemption takes effect, the setup success rate and average rate of services with high ARP priorities increase but services of UEs with low ARP priorities will be preempted, causing the following KPIs to increase:
    - **SgNB-Triggered SgNB Abnormal Release Rate (CU)**, which indicates the abnormal SgNB releases triggered by the SgNB in NSA DC scenarios
    - **SgNB Abnormal Release Rate (CU)**, which indicates the abnormal SgNB release rate in NSA DC scenarios
- The way the gNodeB performs UE preemption handling will increase the amount of radio signaling because of the following actions: The gNodeB sets

up a temporary RRC connection for any UE requesting access to the gNodeB in order to obtain its ARP attributes; the gNodeB releases temporary RRC connections for those UEs that eventually encounter preemption failures; the gNodeB releases RRC connections for UEs whose resources are preempted.

- In scenarios where the cell/network slice load is relatively high, selecting the **REJECT_WAIT_TIME_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter leads to an increase in service drop rates (**Service Call Drop Rate (CU)** and **Service Call Drop Rate (CU, Inactive)**), and a decrease in the following: number of RRC connection setup attempts (**N.RRC.SetupReq.Att** + **N.RRC.ResumeReq.Att**), number of bearer setup attempts (**N.QosFlow.Est.Att** + **N.QosFlow.Resume.Att**), and number of online UEs (**N.User.RRCConn.Max**, **N.User.RRCConn.Avg**, and **N.User.NsaDc.PSCell.Avg**).
- Making the following configuration changes during busy hours will lead to excessive service admission within a period of time. As a result, the gains brought by service admission and service preemption decrease. The impact will disappear as these admitted services are gradually released.
  - Turning on the service admission switch. When the service admission switch is turned on in a cell, the statistics of only newly admitted services, rather than the previously admitted services, are collected in the cell and network slice groups for the cell.
  - Running the **ADD NRDUCELLRES** command. When this command is executed, the statistics of the previously admitted services are still collected in the original network slice group, but not in the new network slice group.
  - Running the **RMV NRDUCELLRES** command. When this command is executed, the statistics of the services that have been admitted to the removed network slice group are not collected in other network slice groups.
- If the UE preemption or service preemption function takes effect and the resources of an SA UE are preempted, the *deprioritisationReq* IE will be included the RRCRelease message during the RRC connection release, indicating that the UE should consider the priority of its serving frequency to be the lowest. In this case, if no other NR frequencies are available, the UE will reselect to an LTE cell.
- After "no resource reservation for emergency call UEs" is enabled, the access success rate (**RRC Setup Success Rate (CU)**) increases for common UEs and decreases for emergency call UEs if the UE number resources of a baseband processing unit are limited.
- When the **NRDUCellRac**.***UeNumPreemptionRange*** parameter is set to **INTER_OPERATOR** or **ALL_OPERATOR** for a cell, and the number of UEs in the cell has reached the upper limit, the following impacts may be introduced: (1) The access success rate (**RRC Setup Success Rate (CU)**) increases for high-priority UEs and decreases for low-priority UEs. (2) Low-priority UEs that have accessed the cell may not be able to continue to perform services in the cell.
- When the **NRCellPri**.***VoiceUeNumOfs*** parameter is set to a non-zero value, the access success rate of non-voice service UEs decreases and the service drop rate increases. Although the access success rate of voice service UEs increases, the values of the following indicators change when the cell or network slice is heavily loaded:

- The number of RRC connection setup attempts (**N.RRC.SetupReq.Att** + **N.RRC.ResumeReq.Att**) decreases.
- The number of bearer setup attempts (**N.QosFlow.Est.Att** + **N.QosFlow.Resume.Att**) decreases.
- The values of the following counters related to the number of online UEs decrease:
  - **N.User.RRCConn.Max**, which indicates the maximum number of UEs in RRC_CONNECTED mode in a cell
  - **N.User.RRCConn.Avg**, which indicates the average number of UEs in RRC_CONNECTED mode in a cell
  - **N.User.NsaDc.PSCell.Avg**, which indicates the average number of LTE-NR NSA DC users using the current cell as PSCell
- Service drop rates (such as **Service Call Drop Rate (CU)** and **Service Call Drop Rate (CU, Inactive)**) increase.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	gNodeB Multi Operator Sharing Mode	**gNBSharingMode**.***gNBMultiOpSharingMode***	*Multi-Operator Sharing*	The **NRDUCellRac**.***UeNumPreemptionRange*** parameter can be set to **INTER_OPERATOR** or **ALL_OPERATOR** only when the **gNBSharingMode**.***gNBMultiOpSharingMode*** parameter is set to **SHARED_FREQ**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Voice UE Number Offset	**NRCellPri**.***VoiceUeNumOfs***	*Admission Control*	The **NRCellPri**.***TempAdmUeNumThld*** parameter can be set to a value other than 20 only when the **NRCellPri**.***VoiceUeNumOfs*** parameter is set to a value other than 0.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink with independent SUL deployment	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Service admission and service preemption cannot be enabled in SUL cells.
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Service admission and service preemption cannot be enabled in hyper cells.
Low-frequency TDD	LTE FDD and NR Uplink Spectrum Sharing	**LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	When the switch for LTE FDD and NR Uplink Spectrum Sharing is turned on, service admission and service preemption cannot be enabled.
Low-frequency TDD	LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	When the switch for LTE TDD and NR spectrum sharing is turned on, service admission and service preemption cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When user-rate-based dynamic network slice resource management is enabled, service admission cannot be enabled.
Low-frequency TDD	User-rate-based dynamic QCI resource management	**DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When user-rate-based dynamic QCI resource management is enabled, service admission cannot be enabled.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA private line service assurance (TDD)	**RRC_SETUP_REASON_ADMISSION_SW** and **PRIVATE_LINE_SW** options of the **NRCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When both the switch for FWA private line service assurance and that for UE-number-based preemption are turned on in a cell, the preempting UEs, for which the cause value of RRC connection setup is not **Emergency**, **highPriorityAccess**, or **mo-VoiceCall**, use temporary resources of private line UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA private line service assurance (TDD)	**RRC_SETUP_REASON_ADMISSION_SW** and **PRIVATE_LINE_SW** options of the **NRCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When the **NRDUCellRac**.***SpecServPviMode*** parameter is set to **PVI_FOLLOW_ARP**, resources of bearers for FWA private line UEs may be preempted by the bearers of other UEs of higher ARP priorities if these UEs' bearers have relatively low ARP priorities and can be preempted. It is recommended that such UEs be configured with high ARP priorities on the core network and their resources not be preempted when FWA private line service assurance is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When the **NRDUCellRac**.***SpecServPviMode*** parameter is set to **PVI_FOLLOW_ARP**, resources of bearers for UEs running low latency and high reliability services may be preempted by the bearers of other UEs of higher ARP priorities if these UEs' bearers have relatively low ARP priorities and can be preempted. It is recommended that such UEs be configured with high ARP priorities on the core network and their resources not be preempted when the low latency and high reliability function is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	When the **NRDUCellRac**.***SpecServPviMode*** parameter is set to **PVI_FOLLOW_ARP**, resources of bearers for VoNR UEs may be preempted by the bearers of other UEs of higher ARP priorities if these UEs' bearers have relatively low ARP priorities and can be preempted. It is recommended that such UEs be configured with high ARP priorities on the core network and their resources not be preempted when basic VoNR functions are enabled.
High-frequency TDD	None	None	None	None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

The preemption capability and vulnerability must be specified in the core network for services with different priorities.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-3 and **Table 4-4** describe the parameters used for function activation and optimization, respectively.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
RAC Algorithm Switch	**NRCellAlgoSwitch**.***RacAlgoSwitch***	USER_NUM_PREEMPTION_SW	It is recommended that this option be selected for heavily loaded cells.
RAC Algorithm Switch	**NRDUCellAlgoSwitch**.***RacAlgoSwitch***	QOS_FLOW_ADMISSION_SW	It is recommended that this option be selected for heavily loaded cells.
RAC Algorithm Switch	**NRDUCellAlgoSwitch**.***RacAlgoSwitch***	QOS_FLOW_PREEMPTION_SW	It is recommended that this option be selected for heavily loaded cells where the **QOS_FLOW_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected.

Parameter Name	Parameter ID	Option	Setting Notes
Network Slice Algorithm Switch	**NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch***	RB_DYNAMIC_CONTROL_SW	It is recommended that this option be selected for heavily loaded cells.
RAC Algorithm Switch	**NRCellAlgoSwitch**.***RacAlgoSwitch***	REJECT_WAIT_TIME_SW	It is recommended that this option be selected for heavily loaded cells.
RAC Algorithm Switch	**NRDUCellAlgoSwitch**.***RacAlgoSwitch***	RES_UNRESERVED_FOR_EMC_SW	It is recommended that this option be selected in cells where the UE number resources of the baseband processing unit are limited and no special guarantee is required for emergency call UEs.
UE Number Preemption Range	**NRDUCellRac**.***UeNumPreemptionRange***	None	Set this parameter based on the network plan.
Voice UE Number Offset	**NRCellPri**.***VoiceUeNumOfs***	None	Set this parameter based on the network plan.
Temp Admission Timer	**NRCellPri**.***TempAdmTimer***	None	Set this parameter based on the network plan.
Voice UE Number Offset	**NRCellPri**.***TempAdmUeNumThld***	None	Set this parameter based on the network plan.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
UE-Number-based Admission Threshold	**NRCellPri**.***UserNumAdmThld***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Handover/Reestablishment UE Number Offset	**NRCellPri**.*HoReestUeNumOfs*	Set this parameter based on the network plan.
High-Priority UE Number Offset	**NRCellPri**.*HighPriUeNumOfs*	Set this parameter based on the network plan.
Downlink Guaranteed Rate	**gNBDUMacParamGroup**.*DlGuaranteedRate*	Set this parameter based on the network plan.
Uplink Guaranteed Rate	**gNBDUMacParamGroup**.*UlGuaranteedRate*	Set this parameter based on the network plan.
Downlink PRB Usage Threshold	**NRDUCellRac**.*DlPrbUsageThld*	The default value is recommended.
Uplink PRB Usage Threshold	**NRDUCellRac**.*UlPrbUsageThld*	The default value is recommended.
Uplink Admission Threshold Factor	**NRDUCellRac**.*UlAdmissionThldFactor*	The default value is recommended.
Downlink Admission Threshold Factor	**NRDUCellRac**.*DlAdmissionThldFactor*	The default value is recommended.
RAC Switch	**gNBOperatorQciParam**.*RacSwitch*	The default value is recommended.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Activation command examples for UE admission control

```
//Modifying the UserNumAdmThld, HoReestUeNumOfs, and HighPriUeNumOfs parameters
MOD NRCELLPRI: NrCellId=7, UserNumAdmThld=40, HoReestUeNumOfs=20, HighPriUeNumOfs=20;
//Turning on the UE-number-based preemption algorithm switch
MOD NRCELLALGOSWITCH: NrCellId=7, RacAlgoSwitch=USER_NUM_PREEMPTION_SW-1;
//Modifying the range of UEs for UE-number-based preemption to INTER_OPERATOR
MOD NRDUCELLRAC: NrDuCellId=7, UeNumPreemptionRange=INTER_OPERATOR;
//Modifying the range of UEs for UE-number-based preemption to ALL_OPERATOR
MOD NRDUCELLRAC: NrDuCellId=7, UeNumPreemptionRange=ALL_OPERATOR;
//Turning on the switch for "no resource reservation for emergency call UEs"
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, RacAlgoSwitch=RES_UNRESERVED_FOR_EMC_SW-1;
//Selecting the HIGH_ACCESS_PRIORITY option of the gNBOperatorQciParam.RacSwitch parameter
corresponding to the bearer with a QCI of 8 for a UE to prevent the UE from being preempted
MOD GNBOPERATORQCIPARAM: OperatorId=0, Qci=8, RacSwitch=HIGH_ACCESS_PRIORITY-1;
//Enabling admission guarantee for voice service UEs
MOD NRCELLPRI: NrCellId=7, VoiceUeNumOfs=20;
//(Optional) Configuring the number of temporarily admitted UEs in admission guarantee for voice service
UEs
MOD NRCELLPRI: NrCellId=7, TempAdmUeNumThld=40;
//(Optional) Configuring the temporary admission timer in admission guarantee for voice service UEs
MOD NRCELLPRI: NrCellId=7, TempAdmTimer=5;
```

Activation command examples for service admission control

```
//Turning on the service admission algorithm switch and service preemption algorithm switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=7,
RacAlgoSwitch=QOS_FLOW_ADMISSION_SW-1&QOS_FLOW_PREEMPTION_SW-1;
//Turning on the dynamic RB control switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Turning on the rejection waiting time switch
MOD NRCELLALGOSWITCH: NrCellId=7, RacAlgoSwitch=REJECT_WAIT_TIME_SW-1;
//Adding a gNodeB MAC parameter group
ADD GNBDUMACPARAMGROUP: MacParamGroupId=1, DlGuaranteedRate=10, UlGuaranteedRate=10;
//Associating the NR DU cell with the gNodeB MAC parameter group
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=8, MacParamGroupId=1;
//Adjusting the uplink PRB usage threshold, downlink PRB usage threshold, uplink admission threshold
factor, and downlink admission threshold factor for service admission
MOD NRDUCELLRAC: NrDuCellId=7, DlPrbUsageThld=99, UlPrbUsageThld=98, DlAdmissionThldFactor=0,
UlAdmissionThldFactor=0;
//Selecting the HIGH_ACCESS_PRIORITY option of the gNBOperatorQciParam.RacSwitch parameter
corresponding to the bearer with a QCI of 8 for a service to prevent the service from being preempted
MOD GNBOPERATORQCIPARAM: OperatorId=0, Qci=8, RacSwitch=HIGH_ACCESS_PRIORITY-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the UE-number-based preemption switch
MOD NRCELLALGOSWITCH: NrCellId=7, RacAlgoSwitch=USER_NUM_PREEMPTION_SW-0;
//Turning off the service admission algorithm switch and service preemption algorithm switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=7,
RacAlgoSwitch=QOS_FLOW_ADMISSION_SW-0&QOS_FLOW_PREEMPTION_SW-0;
//Modifying the range of UEs for UE-number-based preemption to INTRA_OPERATOR
MOD NRDUCELLRAC: NrDuCellId=7, UeNumPreemptionRange=INTRA_OPERATOR;
//Disabling admission guarantee for voice service UEs
MOD NRCELLPRI: NrCellId=7, VoiceUeNumOfs=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Scenario 1: UE Preemption

The following assumptions apply: A cell supports up to 400 UEs; the UE-number-based preemption switch is turned on; the number of admitted UEs has reached the upper limit; high-priority UEs can access the cell by preempting resources of low-priority UEs. The verification procedure is as follows:

Step 1 Start NG signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **NG Interface Trace**.

Step 2 Select the **USER_NUM_PREEMPTION_SW** option of the **NRCellAlgoSwitch**.***RacAlgoSwitch*** parameter.

Step 3 Enable a UE that meets the following requirements to access the network: The ARP value is 7 and the *Pre-emption Vulnerability* IE is set to **pre-emptable** for the default bearer of the UE.

Step 4 Enable 399 UEs that are served by the same operator as the UE in **Step 3** to access the network. For the default bearers of these UEs, the ARP value is 6 and the *Pre-emption Vulnerability* IE is set to **pre-emptable**.

Step 5 Enable another UE that is served by the same operator as the UE in **Step 3** to access the network. For the default bearer of the new UE, the ARP value is 1 and the *Pre-emption Capability* IE is set to **may trigger pre-emption**.

----End

Expected result: The new UE successfully accesses the network, and the resources of the UE whose default bearer corresponds to the ARP of 7 are preempted.

- In SA networking, the cause value in the UE CONTEXT RELEASE REQUEST message traced over the NG interface is **Radio Network: Release due to pre-emption**.
- In NSA networking, the cause value in the SgNB Release Require message traced over the NG interface is **Radio Network: Reduce Load** for the UE whose resources are preempted.

Scenario 2: Service Admission

Assume that GBR services with different QCIs are established. The verification procedure is as follows:

Step 1 Start NG signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **NG Interface Trace**.

Step 2 Select the **QOS_FLOW_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter.

Step 3 Enable UEs to access the network and set up services with a QCI of 4 on some UEs.

Step 4 Set up services with a QCI of 3 on some UEs. Simulate a scenario where new services cannot be admitted based on the PRB usage, and set up services with a QCI of 2.

----End

Expected result: The services with the QCI of 2 fail to be admitted.

Scenario 3: Service Preemption

The following assumptions apply: GBR services with different QCIs are set up; the GBR service with the QCI of 4 has a higher ARP value than that with the QCI of 3 (a larger ARP value indicates a lower priority); the ARP value of the GBR service with the QCI of 3 is 7; and both GBR services with QCIs 3 and 4 can preempt resources of other services and their resources can also be preempted. The ARP value and ARP attributes can be observed through the PDU SESSION RESOURCE SETUP REQUEST message traced over the NG interface. The verification procedure is as follows:

Step 1 Start NG signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **NG Interface Trace**.

Step 2 Select the **QOS_FLOW_ADMISSION_SW** and **QOS_FLOW_PREEMPTION_SW** options of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter.

Step 3 Enable UEs to access the network and set up low-priority services (for example, services with a QCI of 4) on some UEs.

Step 4 Set up high-priority services (for example, services with a QCI of 3) on some UEs. Simulate a scenario where new services cannot be admitted based on the PRB usage. Based on preemption principles, new high-priority services (with a QCI of 3 and meeting the preempting conditions) can be admitted by preempting the resources of low-priority services.

----End

Expected result: The services with the QCI of 3 successfully preempt the resources of services with the QCI of 4. That is, the resources of services with the QCI of 4 are released due to preemption.

- In SA networking, the cause value in the UE CONTEXT RELEASE REQUEST message traced over the NG interface is **Radio Network: Release due to pre-emption**.
- In NSA networking, the cause value in the SgNB Release Require message traced over the NG interface is **Radio Network: Reduce Load** for the UE whose resources are preempted.

4.4.3 Network Monitoring

When UE preemption is activated, an RRC connection is set up, and preemption is then triggered. In this situation, the following counters are measured:

- **N.UserSpec.Preempt.Att**, which measures the number of times preemptions are triggered due to the insufficiency of UE number specifications

- **N.UserLic.Preempt.Att**, which measures the number of times preemptions are triggered due to the insufficiency of UE number licenses in a base station.

If admission or preemption of a UE fails, the RRC connection of the UE is released. In this situation, the following counters are measured:

- **N.UECntx.FailEst.NoRadioRes**, which measures the number of SA UE context setup failures because admission or preemption fails due to the insufficiency of UE number specifications
- Cell-level counter **N.RRCConn.UserLic.Limit.Num** and PLMN-level counter **N.RRCConn.UserLic.Limit.Num.PLMN**, which measure the number of SA UE context setup failures because admission or preemption fails as a result of limited licensed number of RRC_CONNECTED UEs
- **N.NsaDc.SgNB.Add.Fail.Radio.License**, which measures the number of SgNB addition failures for NSA UEs because admission or preemption fails due to the insufficiency of UE number specifications

If preemption succeeds, the RRC connection of the UE whose resources are preempted is released. In this situation, the following counters are measured:

- **N.UECntx.AbnormRel.UserSpec.Preempt** and **N.UECntx.AbnormRel.UserLic.Preempt**, which are used to measure the number of RRC connection releases
- **N.UserSpec.Preempt.Succ** and **N.UserLic.Preempt.Succ**, which are used to measure the number of successful preemptions

The number of admission and preemption failures due to insufficient PRB resources in a network slice/cell is measured using the following counters:

- **N.QosFlow.FailEst.ResNotAvailForNs**
- **N.QosFlow.FailEst.NoRadioRes**
- **N.NsaDc.SgNB.AbnormRel.Radio**

If service preemption succeeds, the number of service and RRC connection releases for the UE whose resources are preempted is measured using the following counters:

- **N.QosFlow.AbnormRel.RNL.Preempt** and **N.NsaDc.DRB.AbnormRel.RNL.Preempt**, which are used to measure the number of service releases
- **N.UECntx.AbnormRel.RNL.Preempt** and **N.NsaDc.SgNB.AbnormRel.RNL.Preempt**, which are used to measure the number of RRC connection releases

Log in to the MAE, and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type** > **NR** > **Cell Performance Monitoring** > **Admission Control Monitoring**. In the displayed dialog box, specify related parameters to start admission control monitoring.

- To observe the uplink and downlink PRB quotas of admitted services in a cell, check the values of **Cell Admission QoS Flow UL PRB Quota** and **Cell Admission QoS Flow DL PRB Quota**.
- To observe the uplink and downlink PRB quotas of admitted services in a slice resource group, check the values of the following counters:

- **Network Slice Group ID 0 Admission QoS Flow UL PRB Quota** to **Network Slice Group ID 5 Admission QoS Flow UL PRB Quota**
- **Network Slice Group ID 0 Admission QoS Flow DL PRB Quota** to **Network Slice Group ID 5 Admission QoS Flow DL PRB Quota**

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *FWA*
- *URLLC*
- *VoNR*
- *Network Slicing*
- *Hyper Cell*
- *Super Uplink*
- *Transmission Resource Management*
- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- *Multi-Operator Sharing*
- *QCI-based Resource Control*
- *BBU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*